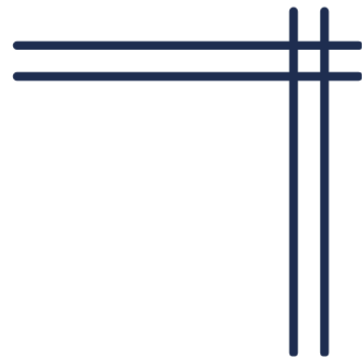




COLLEGE OF
ENGINEERING
POONJAR

(Managed by IHRD, Govt. of Kerala)



GROUP 4

PROJECT

SENTINEL_FUSION

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Functional Requirements

Introduction

Functional requirements define the core functionalities and services that **Sentinel Fusion** is expected to perform. These requirements describe the interactions between users and the system, ensuring that digital evidence is securely managed, intelligently analyzed, and presented in a manner that supports efficient forensic investigations.

The following functional requirements have been identified for the proposed system.

User Authentication and Authorization

Description

The system shall provide secure authentication and authorization mechanisms to ensure that only authorized users can access investigation data.

Functional Requirements

- Users shall be able to log in using secure credentials.
- The system shall support Multi-Factor Authentication (MFA).
- Role-Based Access Control (RBAC) shall be implemented.
- Administrators shall manage user accounts and permissions.
- User sessions shall be securely maintained.

Case Management

Description

The system shall enable investigators to organize digital evidence into individual investigation cases.

Functional Requirements

- Create new investigation cases.
- Update case information.
- Assign investigators to cases.
- Archive completed investigations.
- Maintain case history.

Evidence Ingestion and Management

Description

The system shall securely collect, verify, and store multiple types of digital evidence.

Functional Requirements

- Upload images, videos, audio files, documents, chat exports, and ZIP archives.
- Support bulk evidence upload.
- Generate SHA-256 hash values for integrity verification.
- Record chain of custody information.
- Store evidence securely in object storage.

Metadata Extraction

Description

The system shall automatically extract metadata from uploaded digital evidence.

Functional Requirements

- Extract timestamps.
- Extract GPS coordinates.
- Extract device information.
- Extract camera details.
- Extract document properties.
- Store extracted metadata in the database.

AI-Based Evidence Analysis

Description

The system shall use Artificial Intelligence to automate evidence analysis and extract meaningful information.

Functional Requirements

- Perform OCR on images and scanned documents.
- Convert audio recordings into text using Speech-to-Text.

- Analyze images and videos using Computer Vision.
- Extract named entities such as persons, locations, organizations, and devices.
- Generate AI-assisted summaries of evidence.

Evidence Correlation and Relationship Analysis

Description

The system shall identify relationships between entities across multiple evidence sources.

Functional Requirements

- Correlate evidence from different files.
- Identify connections between people, devices, and locations.
- Build an interactive relationship graph.
- Highlight hidden relationships.

Timeline Reconstruction

Description

The system shall automatically reconstruct investigation timelines.

Functional Requirements

- Arrange events chronologically.
- Combine timestamps from multiple evidence sources.
- Display an interactive timeline.
- Allow filtering by date and event type.

Synthetic Media Detection

Description

The system shall assist investigators in identifying manipulated digital content.

Functional Requirements

- Detect AI-generated images.

- Detect manipulated videos.
- Detect cloned or synthetic voices.
- Provide confidence scores for authenticity analysis.

Investigation Dashboard

Description

The system shall provide investigators with a centralized dashboard to monitor case progress.

Functional Requirements

- Display case overview.
- Display evidence statistics.
- Show AI findings.
- Visualize relationship graphs.
- Display investigation timelines.
- Present risk indicators and alerts.

Report Generation

Description

The system shall generate comprehensive investigation reports.

Functional Requirements

- Generate AI-assisted investigation summaries.
- Include evidence inventory.
- Include timeline reconstruction.
- Include relationship analysis.
- Export reports in PDF, DOCX, and JSON formats.

Audit Logging and Evidence Integrity

Description

The system shall maintain complete traceability of all investigation activities.

Functional Requirements

- Record every user action.
- Maintain immutable audit logs.
- Verify evidence integrity using stored hash values.
- Track chain of custody throughout the investigation.

Summary of Functional Requirements

Module	Primary Function
User Authentication	Secure login and access control
Case Management	Organize and manage investigation cases
Evidence Management	Upload, verify, and securely store digital evidence
Metadata Extraction	Automatically extract forensic metadata
AI-Based Analysis	OCR, speech recognition, vision analysis, and AI summaries
Evidence Correlation	Discover relationships among entities
Timeline Reconstruction	Build chronological event timelines
Synthetic Media Detection	Identify manipulated and AI-generated media
Investigation Dashboard	Visualize evidence, AI findings, and case progress
Report Generation	Create investigation reports in multiple formats
Audit Logging	Maintain evidence integrity and accountability

Non-Functional Requirements

Introduction

Non-functional requirements define the quality attributes and operational characteristics of **Sentinel Fusion**. These requirements ensure that the system is secure, reliable, scalable, maintainable, and capable of delivering consistent performance while handling sensitive digital evidence and supporting AI-assisted forensic investigations.

Performance

The system should provide fast and efficient processing of digital evidence to support timely investigations.

Requirements

- Dashboard response time should be less than **2 seconds** under normal operating conditions.
- AI-generated investigation summaries should be produced within **30 seconds** for standard evidence sets.
- The system should support concurrent access by multiple authorized users.
- Large multimedia files should be uploaded and processed efficiently.
- Background AI tasks should not affect the responsiveness of the user interface.

Security

As Sentinel Fusion handles sensitive forensic evidence, strong security measures must be implemented.

Requirements

- All communication shall be secured using **TLS 1.3**.
- Sensitive information shall be encrypted using **AES-256** encryption.
- Role-Based Access Control (RBAC) shall restrict user permissions.
- Multi-Factor Authentication (MFA) shall be supported.
- Evidence integrity shall be verified using **SHA-256** hash values.
- Immutable audit logs shall record all user activities.

- Secure backup and recovery mechanisms shall be implemented.

Reliability

The system should provide consistent and dependable operation throughout the investigation process.

Requirements

- The system should achieve an availability of **99.9%** during normal operation.
- Uploaded evidence shall not be lost due to system failures.
- Automatic recovery mechanisms shall restore interrupted background processes.
- Data consistency shall be maintained across all system components.

Scalability

The platform should be capable of handling increasing workloads as investigation data grows.

Requirements

- Support storage and management of **10,000 or more** evidence files.
- Support horizontal scaling of backend services.
- Enable distributed AI processing using worker nodes.
- Allow future integration of additional AI models and forensic tools.

Usability

The system should provide an intuitive interface that minimizes the learning curve for investigators.

Requirements

- Provide a clean and responsive dashboard.
- Ensure easy navigation between investigation modules.
- Display interactive graphs, timelines, and visualizations.
- Present AI-generated insights in an understandable and explainable format.

- Support modern web browsers and different screen resolutions.

Maintainability

The software should be easy to update, test, and extend.

Requirements

- Adopt a modular software architecture.
- Follow coding standards and documentation practices.
- Support independent updates of AI modules.
- Maintain comprehensive API documentation.
- Enable automated testing and continuous integration.

Availability

The system should remain accessible whenever authorized users require it.

Requirements

- Ensure continuous access to investigation data.
- Minimize downtime during maintenance activities.
- Support automatic service restart after unexpected failures.

Compatibility

The system should operate seamlessly across different environments.

Requirements

- Support major web browsers such as Chrome, Firefox, Edge, and Safari.
- Accept common digital evidence formats, including images, videos, audio, documents, and chat exports.
- Be deployable on cloud platforms as well as on-premises infrastructure.

Summary of Non-Functional Requirements

Quality Attribute Requirement

Performance	Fast response, efficient AI processing, concurrent user support
Security	TLS 1.3, AES-256, RBAC, MFA, SHA-256, immutable audit logs
Reliability	High availability, fault tolerance, automatic recovery
Scalability	Support for large evidence volumes and distributed processing
Usability	User-friendly dashboard and interactive visualizations
Maintainability	Modular architecture, documentation, automated testing
Availability	Continuous service with minimal downtime
Compatibility	Cross-browser support and multiple evidence formats

Literature Survey

Introduction

The rapid advancement of digital technology has significantly increased the volume and complexity of digital evidence generated through smartphones, computers, cloud platforms, social media, and Internet of Things (IoT) devices. Digital forensic investigations rely on the collection, preservation, analysis, and presentation of this evidence to support legal proceedings and cybersecurity incident response. However, the growing diversity of digital evidence has made manual analysis increasingly challenging and time-consuming.

Traditional digital forensic tools are effective in acquiring and preserving digital evidence but often require investigators to manually examine multiple evidence sources, establish relationships, reconstruct timelines, and prepare investigation reports. As investigation datasets continue to grow, these manual processes become less efficient and increase the possibility of overlooking critical evidence.

Recent developments in Artificial Intelligence (AI), Machine Learning (ML), Computer Vision, Natural Language Processing (NLP), Graph Databases, and Large Language Models (LLMs) have opened new opportunities for enhancing digital forensic investigations. These technologies enable automated metadata extraction, intelligent content analysis, relationship discovery, semantic search, and explainable report generation, thereby improving investigator productivity and reducing analysis time.

This literature survey reviews existing research, commercial forensic solutions, and enabling technologies relevant to AI-assisted digital forensic investigations. It also identifies the limitations of current approaches, highlights existing research gaps, and explains how the proposed **Sentinel Fusion** platform addresses these challenges through an integrated, AI-powered digital intelligence fusion framework.

Research Papers

For academic quality, review **6 research papers**. Use the same format for each paper.

Paper 1

Title: *Artificial Intelligence in Digital Forensics: A Review*

Objective

The paper explores how Artificial Intelligence techniques can improve digital forensic investigations by automating evidence analysis, pattern recognition, and decision support.

Methodology

The authors review AI techniques including Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing used in forensic investigations.

Key Findings

- AI reduces manual investigation effort.
- Machine Learning improves evidence classification.
- Automated analysis accelerates investigations.
- AI assists investigators in handling large datasets.

Limitations

- Limited explainability of AI models.
- Dependence on high-quality datasets.
- Human verification remains necessary.

Relevance to Sentinel Fusion

This research supports the integration of AI-assisted evidence analysis within Sentinel Fusion while reinforcing the importance of maintaining human oversight and explainable decision-making.

Paper 2

Title: *Deep Learning for Digital Forensics: A Survey*

Objective

This paper examines the application of deep learning techniques in digital forensic investigations, particularly for image analysis, malware detection, multimedia forensics, and evidence classification.

Methodology

The authors review various deep learning models, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), used in forensic applications.

Key Findings

- Deep learning improves image and video analysis accuracy.
- Automated feature extraction reduces manual effort.
- AI models can process large volumes of multimedia evidence efficiently.
- Better performance than traditional machine learning techniques for complex forensic tasks.

Limitations

- Requires large labeled datasets for training.
- High computational cost.
- Limited interpretability of model predictions.

Relevance to Sentinel Fusion

This research supports the use of AI-powered vision models for multimedia evidence analysis and automated content classification within Sentinel Fusion.

Paper 3

Title: *Knowledge Graphs for Digital Investigation and Intelligence Analysis*

Objective

The paper investigates the use of knowledge graphs to represent relationships among digital entities such as people, devices, locations, and online accounts.

Methodology

The study models investigation data as interconnected graph structures and evaluates graph-based querying and visualization techniques.

Key Findings

- Graph databases simplify relationship analysis.
- Hidden connections between entities become easier to identify.
- Graph visualization improves investigator understanding.

Limitations

- Graph construction requires accurate entity extraction.
- Performance decreases with poorly structured data.
- Integration with forensic tools remains limited.

Relevance to Sentinel Fusion

This paper forms the foundation for implementing Neo4j-based relationship mapping and intelligence fusion within Sentinel Fusion.

Paper 4

Title: *Explainable Artificial Intelligence for Trustworthy Decision Support Systems*

Objective

The research focuses on improving transparency and trust in AI systems by providing explanations for AI-generated decisions.

Methodology

The authors evaluate different Explainable AI (XAI) techniques for presenting confidence scores, reasoning paths, and supporting evidence.

Key Findings

- Explainability improves user trust.
- Confidence scores help validate AI decisions.
- Human oversight remains essential for critical applications.

Limitations

- Explainability may increase computational complexity.
- No single XAI approach is suitable for all AI models.

Relevance to Sentinel Fusion

This paper supports Sentinel Fusion's human-in-the-loop approach, where every AI-generated insight is accompanied by supporting evidence and confidence scores.

Paper 5

Title: *Deepfake Detection Using Artificial Intelligence: A Review*

Objective

The paper reviews AI techniques for detecting manipulated images, videos, and synthetic audio.

Methodology

The authors analyze deep learning models developed for detecting facial manipulation, voice cloning, and AI-generated multimedia.

Key Findings

- Deep learning significantly improves deepfake detection.
- Multimodal analysis enhances detection accuracy.
- AI-generated media presents a growing challenge for digital investigations.

Limitations

- Detection accuracy decreases as deepfake technology evolves.
- Frequent retraining of models is required.

Relevance to Sentinel Fusion

This research supports the inclusion of synthetic media detection capabilities to assist investigators in verifying the authenticity of digital evidence.

Paper 6

Title: *Artificial Intelligence for Automated Digital Evidence Analysis*

Objective

This paper explores AI-assisted automation for digital evidence processing, including metadata extraction, document analysis, and report generation.

Methodology

The authors combine Natural Language Processing, Computer Vision, and metadata analysis to automate forensic workflows.

Key Findings

- AI reduces investigation time.
- Automated report generation improves documentation.
- Combining multiple AI techniques increases investigation efficiency.

Limitations

- AI cannot replace expert investigators.
- False positives require manual verification.

Relevance to Sentinel Fusion

This paper directly aligns with Sentinel Fusion's objective of providing AI-assisted evidence analysis while ensuring that investigators retain full control over all investigative decisions.

Existing Solutions

Several commercial and open-source digital forensic tools are widely used for collecting, analyzing, and managing digital evidence. These tools provide essential capabilities for forensic investigations but often focus on specific aspects of evidence analysis rather than offering a unified AI-assisted investigation platform.

Autopsy

Description

Autopsy is a popular open-source digital forensic platform used for analyzing hard drives, mobile devices, and file systems. It provides features such as keyword searching, metadata extraction, file recovery, timeline analysis, and report generation.

Features

- File system analysis
- Metadata extraction
- Timeline analysis
- Keyword search
- Report generation

Limitations

- Limited AI-powered automation.
- No integrated relationship graph analysis.
- Manual correlation of evidence is required.

FTK (Forensic Toolkit)

Description

FTK is a commercial digital forensic tool developed by AccessData. It is widely used for evidence acquisition, indexing, email analysis, and password recovery.

Features

- High-speed indexing
- Email analysis
- File recovery

- Evidence processing
- Report generation

Limitations

- Limited explainable AI capabilities.
- Requires significant manual investigation.
- Relationship visualization is limited.

Magnet AXIOM

Description

Magnet AXIOM is a comprehensive forensic investigation platform designed to collect and analyze evidence from computers, mobile devices, and cloud services.

Features

- Cloud evidence acquisition
- Mobile device analysis
- Chat and social media analysis
- Timeline reconstruction
- Artifact recovery

Limitations

- AI-assisted evidence correlation is limited.
- No integrated deepfake detection.
- Complex investigations still require extensive manual analysis.

Cellebrite UFED

Description

Cellebrite UFED is widely used by law enforcement agencies for extracting and analyzing data from mobile devices.

Features

- Mobile data extraction
- Device unlocking support

- Application data recovery
- Multimedia analysis

Limitations

- Primarily focused on mobile devices.
- Limited support for integrating multiple evidence sources.
- Does not provide comprehensive AI-driven intelligence fusion.

Belkasoft Evidence Center

Description

Belkasoft Evidence Center is a digital forensic solution used for analyzing computers, smartphones, cloud storage, and memory dumps.

Features

- Evidence acquisition
- Memory analysis
- Browser history analysis
- Cloud evidence support
- Timeline generation

Limitations

- Limited AI-powered automation.
- Manual interpretation of investigation results.
- Limited graph-based intelligence analysis.

Comparison of Existing Solutions

Tool	Strengths	Limitations
Autopsy	Open-source, metadata extraction, timeline analysis	Limited AI automation and relationship analysis
FTK	Fast indexing, email analysis, evidence processing	Limited explainable AI and graph visualization

Tool	Strengths	Limitations
Magnet AXIOM	Cloud and mobile forensics, artifact recovery	Limited AI-driven intelligence fusion
Cellebrite UFED	Advanced mobile device forensics	Focused mainly on mobile evidence
Belkasoft Evidence Center	Multi-source evidence analysis	Limited AI automation and manual interpretation

Summary

Existing forensic tools provide reliable evidence acquisition and analysis capabilities but generally require investigators to manually correlate evidence, identify relationships, and interpret findings. They also offer limited support for AI-assisted decision-making, explainable analytics, and unified intelligence fusion.

Sentinel Fusion addresses these limitations by combining AI-powered evidence analysis, graph-based relationship mapping, timeline reconstruction, synthetic media detection, and explainable investigation support within a single integrated platform

Technologies Used

The development of modern digital forensic systems relies on advanced technologies that enable efficient evidence processing, intelligent analysis, secure storage, and meaningful visualization. Sentinel Fusion integrates several emerging technologies to automate forensic investigations while ensuring transparency, security, and human oversight.

Artificial Intelligence (AI)

Artificial Intelligence enables computers to perform tasks that normally require human intelligence. In digital forensics, AI assists investigators by automating evidence analysis, recognizing patterns, identifying suspicious activities, and generating investigation summaries. AI significantly reduces manual effort and improves investigation efficiency.

Applications in Sentinel Fusion

- Intelligent evidence analysis
- Automated investigation summaries
- Risk assessment
- Decision support

Computer Vision

Computer Vision enables machines to interpret and analyze visual information from images and videos. It helps investigators identify objects, faces, scenes, and textual information within multimedia evidence.

Applications in Sentinel Fusion

- Image analysis
- Video content analysis
- Object detection
- Scene understanding
- Evidence categorization

Natural Language Processing (NLP)

Natural Language Processing allows computers to understand and process human language. It enables the extraction of meaningful information from documents, chat messages, emails, and reports.

Applications in Sentinel Fusion

- Chat analysis
- Document processing
- Named Entity Recognition (NER)
- Keyword extraction
- AI-assisted report generation

Optical Character Recognition (OCR)

OCR converts text contained in images, scanned documents, and screenshots into machine-readable text. This allows investigators to search and analyze information that would otherwise require manual transcription.

Applications in Sentinel Fusion

- Text extraction from images
- Document digitization
- Screenshot analysis
- Searchable evidence creation

Speech Recognition

Speech recognition technology converts spoken language into text. It assists investigators in processing interviews, voice recordings, and multimedia evidence.

Applications in Sentinel Fusion

- Audio transcription
- Voice recording analysis
- Searchable audio evidence
- Investigation documentation

6. Graph Databases

Graph databases store information as interconnected nodes and relationships, making them highly effective for representing investigation data. They help investigators visualize connections between people, devices, locations, and digital artifacts.

Applications in Sentinel Fusion

- Relationship mapping
- Network visualization
- Hidden connection discovery
- Intelligence fusion

Large Language Models (LLMs)

Large Language Models are advanced AI models capable of understanding and generating human-like text. They support evidence summarization and explainable investigation reporting.

Applications in Sentinel Fusion

- Investigation summaries
- Explainable AI insights
- Question answering
- Report generation

Metadata Analysis

Metadata provides information about digital files, including timestamps, GPS locations, device details, and document properties. Analyzing metadata helps investigators reconstruct events and verify evidence authenticity.

Applications in Sentinel Fusion

- Timeline reconstruction
- Device identification
- Location tracking
- Evidence verification

Semantic Search and Vector Databases

Semantic search retrieves information based on meaning rather than exact keyword matches. Vector databases store embeddings that enable intelligent search across large collections of digital evidence.

Applications in Sentinel Fusion

- Similar evidence retrieval
- AI-powered search
- Context-aware investigation
- Retrieval-Augmented Generation (RAG)

Cloud Storage and Distributed Computing

Cloud-based object storage and distributed computing allow the system to securely store large multimedia files and process evidence efficiently using multiple AI workers.

Applications in Sentinel Fusion

- Secure evidence storage
- Scalable processing
- High availability
- Large-scale multimedia management

Summary of Technologies

Technology	Purpose in Sentinel Fusion
Artificial Intelligence	Automated evidence analysis and decision support
Computer Vision	Image and video understanding
Natural Language Processing	Text analysis and entity extraction
Optical Character Recognition	Text extraction from images and documents
Speech Recognition	Audio transcription and analysis
Graph Databases	Relationship mapping and intelligence fusion

Technology**Purpose in Sentinel Fusion**

Large Language Models

AI summaries and explainable reporting

Metadata Analysis

Timeline reconstruction and evidence verification

Semantic Search

Intelligent evidence retrieval

Cloud Computing

Scalable storage and distributed AI processing

Limitations of Existing Systems

Although modern digital forensic tools provide reliable capabilities for evidence acquisition, preservation, and analysis, they face several limitations when handling the growing volume and complexity of digital evidence. Most existing solutions require significant manual effort and provide limited support for intelligent evidence correlation and AI-assisted investigation.

The major limitations of existing systems are discussed below.

Manual Evidence Correlation

Most forensic tools analyze evidence individually, requiring investigators to manually correlate information across multiple files, devices, and data sources. This process is time-consuming and increases the possibility of overlooking important relationships.

Limited AI Integration

Many existing forensic platforms provide basic automation but lack advanced AI capabilities such as intelligent evidence summarization, semantic search, automated relationship discovery, and contextual analysis.

Lack of Unified Intelligence Fusion

Digital evidence is often collected from multiple sources, including computers, mobile devices, cloud storage, social media, and communication platforms. Existing tools typically process these sources separately rather than integrating them into a single investigation environment.

Limited Relationship Visualization

Although some tools provide timeline views and artifact listings, they often lack interactive graph-based visualization that helps investigators understand complex relationships among people, devices, locations, and events.

Inadequate Support for Synthetic Media Detection

The increasing use of AI-generated images, videos, and voice recordings presents new challenges for digital investigations. Most traditional forensic tools provide limited or no support for detecting manipulated or synthetic media.

Time-Consuming Investigation Process

Investigators frequently spend considerable time reviewing multimedia files, extracting metadata, preparing reports, and organizing evidence manually. This reduces investigation efficiency, particularly in cases involving large datasets.

Limited Explainability of AI Results

Some AI-enabled systems generate automated outputs without providing sufficient explanations or supporting evidence. This makes it difficult for investigators to verify AI-generated findings and reduces trust in automated analysis.

Scalability Challenges

As the volume of digital evidence continues to grow, many existing systems experience performance issues when processing large datasets or supporting multiple concurrent investigations.

Summary of Limitations

Limitation	Impact on Investigation
Manual evidence correlation	Increased investigation time and workload
Limited AI integration	Reduced automation and slower analysis
Lack of unified intelligence fusion	Fragmented investigation workflow
Limited relationship visualization	Hidden connections may remain undiscovered
Poor synthetic media detection	Difficulty verifying evidence authenticity
Time-consuming manual analysis	Delayed case progression
Limited explainability	Reduced trust in AI-assisted findings
Scalability issues	Performance degradation with large evidence volumes

Research Gap Identified

The literature survey and analysis of existing digital forensic tools reveal that while significant progress has been made in evidence acquisition and forensic analysis, several challenges remain unresolved. Most existing solutions focus on individual forensic tasks rather than providing an integrated platform capable of intelligently correlating evidence from multiple sources.

Current forensic tools largely depend on manual investigation for evidence correlation, relationship discovery, timeline reconstruction, and report preparation. Although Artificial Intelligence has been introduced in certain areas of digital forensics, its adoption remains fragmented, with limited integration across the complete investigation workflow. Furthermore, many existing systems lack explainable AI capabilities, making it difficult for investigators to understand and verify AI-generated insights.

Another significant gap is the limited support for analyzing modern digital evidence, including AI-generated images, deepfake videos, and synthetic audio. As these forms of digital manipulation become increasingly sophisticated, there is a growing need for intelligent systems capable of detecting and assessing the authenticity of digital content.

Additionally, existing forensic platforms often provide limited visualization of relationships among people, devices, locations, and digital artifacts, making it difficult to identify hidden connections within large datasets. The absence of a unified intelligence fusion platform results in fragmented investigations and increased manual effort.

Based on these observations, there is a clear need for a secure, scalable, and AI-assisted digital forensic platform that integrates evidence ingestion, automated analysis, relationship mapping, timeline reconstruction, synthetic media detection, and explainable decision support into a single investigation environment while maintaining human oversight and evidence integrity.

Summary of Research Gaps

Research Gap	Impact
Lack of unified AI-assisted forensic platforms	Fragmented investigation process
Limited automated evidence correlation	Increased manual effort and investigation time

Research Gap	Impact
Insufficient explainable AI	Reduced transparency and trust in AI-generated findings
Limited synthetic media detection	Difficulty verifying manipulated digital evidence
Inadequate relationship visualization	Hidden connections may remain undiscovered
Limited intelligent report generation	Increased documentation effort

Proposed Solution: How Sentinel Fusion Addresses the Research Gap

To overcome the limitations identified in existing digital forensic systems, this project proposes **Sentinel Fusion**, an AI-powered Digital Intelligence Fusion Platform designed to support investigators throughout the entire digital investigation lifecycle. Unlike conventional forensic tools that focus primarily on evidence acquisition and analysis, Sentinel Fusion integrates multiple AI technologies into a unified platform that transforms fragmented digital evidence into actionable investigative intelligence.

The system enables investigators to securely upload and manage evidence from various sources, including images, videos, audio recordings, documents, chat exports, and browser histories. Advanced AI techniques such as Optical Character Recognition (OCR), Computer Vision, Speech-to-Text, and Natural Language Processing are used to automatically extract relevant information from the uploaded evidence. This significantly reduces the manual effort required to examine large volumes of digital data.

Sentinel Fusion further addresses the research gap by introducing **Intelligence Fusion**, where information extracted from multiple evidence sources is automatically correlated to identify relationships between people, devices, locations, and events. A graph database is used to visualize these relationships, enabling investigators to discover hidden connections that may not be apparent through traditional forensic analysis.

To improve investigation efficiency, the platform reconstructs chronological timelines by combining timestamps and metadata from different evidence sources. It also incorporates synthetic media detection capabilities to identify AI-generated or manipulated images, videos, and audio recordings, helping investigators assess the authenticity of digital evidence.

A key feature of Sentinel Fusion is its focus on **Explainable Artificial Intelligence (XAI)**. Rather than producing opaque AI-generated conclusions, the platform provides confidence scores, supporting evidence, and clear explanations for its recommendations. This ensures that investigators remain in control of the decision-making process and can validate AI-assisted findings before taking further action.

By integrating secure evidence management, intelligent analysis, relationship mapping, timeline reconstruction, explainable AI, and automated report generation into a single platform, Sentinel Fusion provides a comprehensive solution that addresses the major challenges identified in current digital forensic systems.

Comparison of Existing Systems and Sentinel Fusion

Feature	Existing Systems	Sentinel Fusion
Evidence Management	Supported	Supported
AI-Based Evidence Analysis	Limited	Comprehensive AI-powered analysis
Automated Evidence Correlation	Limited	Intelligent multi-source correlation
Relationship Graph Visualization	Basic or unavailable	Interactive graph-based intelligence fusion
Timeline Reconstruction	Basic	Automated chronological reconstruction
Deepfake & Synthetic Media Detection	Rarely supported	Integrated AI-powered detection
Explainable AI	Limited	Confidence scores and explainable recommendations
Automated Report Generation	Basic	AI-assisted comprehensive reports
Unified Investigation Platform	Partial	Fully integrated investigation environment

Conclusion

The literature survey demonstrates that existing digital forensic tools provide strong evidence acquisition and analysis capabilities but often lack intelligent automation, integrated evidence correlation, explainable AI, and advanced visualization features. These limitations create opportunities for the development of more comprehensive forensic platforms.

Sentinel Fusion addresses these research gaps by combining Artificial Intelligence, graph analytics, multimedia analysis, timeline reconstruction, and explainable decision support into a secure and scalable platform. The proposed system aims to improve investigation efficiency, reduce manual effort, and assist investigators in deriving meaningful insights from complex digital evidence while maintaining evidence integrity and human oversight.

References

- [1] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet*, 3rd ed. Burlington, MA, USA: Academic Press, 2011.
- [2] B. Carrier, *File System Forensic Analysis*. Boston, MA, USA: Addison-Wesley Professional, 2005.
- [3] National Institute of Standards and Technology (NIST), *Guide to Integrating Forensic Techniques into Incident Response*, Special Publication 800-86, Gaithersburg, MD, USA, 2006.
- [4] National Institute of Standards and Technology (NIST), *Guide to Computer Security Log Management*, Special Publication 800-92, 2006.
- [5] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [6] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2021.
- [7] A. Vaswani *et al.*, "Attention Is All You Need," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [8] OpenAI, "OpenAI API Documentation." Available: <https://platform.openai.com/docs>
- [9] FastAPI, "FastAPI Documentation." Available: <https://fastapi.tiangolo.com/>

- [10] Next.js, "Next.js Documentation." Available: <https://nextjs.org/docs>
- [11] Neo4j, "Neo4j Graph Database Documentation." Available: <https://neo4j.com/docs/>
- [12] PostgreSQL Global Development Group, "PostgreSQL Documentation." Available: <https://www.postgresql.org/docs/>
- [13] Qdrant, "Qdrant Vector Database Documentation." Available: <https://qdrant.tech/documentation/>
- [14] PaddlePaddle, "PaddleOCR Documentation." Available: <https://github.com/PaddlePaddle/PaddleOCR>
- [15] Phil Harvey, "ExifTool Documentation." Available: <https://exiftool.org/>
- [16] OpenAI, "Whisper Speech Recognition." Available: <https://github.com/openai/whisper>
- [17] Docker Inc., "Docker Documentation." Available: <https://docs.docker.com/>
- [18] Kubernetes, "Kubernetes Documentation." Available: <https://kubernetes.io/docs/>
- [19] Apache Software Foundation, "Apache OpenSearch Documentation." Available: <https://opensearch.org/docs/>
- [20] LangGraph Documentation. Available: <https://langchain-ai.github.io/langgraph/>